

GENERAL NOTES:	
1.	LINE SUPPLY AND RETURN AIR DUCTWORK THROUGH FIRST TRANSITION, ELBOW OR FIRST TEN FEET OF EACH UNIT, WHICHEVER IS GREATEST. INTERNAL INSULATION FOR SUPPLY DUCT MUST BE CLOSED-CELL ELASTOMERIC, FIBERGLASS DUCT LINER IN NOT ALLOWED
2.	DUCT SIZES ARE SHOWN AS INSIDE CLEAR DIMENSIONS. WHERE INTERNAL LINING IS CALLED FOR, DIMENSIONS SHALL BE INCREASED BY THE THICKNESS OF THE LINING.
3.	BRANCH ROUND DUCTWORK TAPS LOCATED ABOVE ACCESSIBLE CEILING AND SERVING ONLY ONE DIFFUSER SHALL HAVE SPIN-IN WITH-OUT SCOP.
4.	BRANCH ROUND DUCTWORK LOCATED ABOVE INACCESSIBLE CEILING OR SERVING MORE THAN ONE DIFFUSER SHALL HAVE MANUAL DAMPER OMITTED AND HAVE FACE OPERATED DAMPER IN EACH DIFFUSER UNLESS NOTED OTHERWISE.
5.	REFER TO ARCHITECTURAL REFLECTED CEILING PLANS FOR EXACT LOCATION OF ALL CEILING MOUNTED EQUIPMENT. IF AN ITEM IS NOT SHOWN ON THE R.C.P, PREPARE A DRAWING AND SUBMIT TO THE ARCHITECT FOR APPROVAL.
6.	ALL LAY-IN DIFFUSERS SHALL HAVE FOUR-WAY BLOW UNLESS NOTED OTHERWISE.
7.	MOUNTING FRAME OF CEILING MOUNTED AIR DISTRIBUTION DEVICES SHALL BE COMPATIBLE WITH CEILING TYPE. REFER TO ARCHITECTURAL DRAWINGS FOR CEILING TYPE.
8.	FLEXIBLE DUCT RUNOUTS TO DIFFUSERS SHALL BE INSTALLED FREE OF KINKS AND SAGS. ALL DIFFUSER RUNOUTS SHALL BE SIZED TO MATCH THE INLET OF THE DIFFUSER SERVED.
9.	COORDINATE ALL PENETRATIONS OF STRUCTURAL ELEMENTS WITH STRUCTURAL ENGINEER.
10.	TRANSITION RECTANGULAR DUCTWORK ON THE BOTTOM AND THE SIDES. MAINTAIN DUCTWORK LEVEL AND AS HIGH AS POSSIBLE UNLESS NOTED OTHERWISE.
11.	ALL DUCT CHANGES FROM SQUARE TO ROUND SHALL BE SMOOTH TRANSITIONS. SPIN-INS AT THE END OF CAPPED DUCTS ARE NOT ACCEPTABLE.
12.	PORTIONS OF DUCTWORK OR PIPING VISIBLE THROUGH GRILLES AND REGISTERS IN FINISHED AREAS SHALL BE PAINTED FLAT BLACK.
13.	CONTRACTOR SHALL COORDINATE VOLTAGE AND PHASE OF EACH PIECE OF EQUIPMENT WITH THE ELECTRICAL CONTRACTOR PRIOR TO ORDERING.
14.	MOUNT THERMOSTATS WHERE INDICATED ON PLANS 48" ABOVE FINISHED FLOOR UNLESS NOTED OTHERWISE.
15.	ALL WIRING IN THE CEILING PLENUM SHALL BE PLENUM RATED CABLE OR RUN IN CONDUIT.
16.	CEILING HEIGHTS SHOWN ON ARCHITECTURAL DRAWINGS MUST BE MAINTAINED. NO DUCTWORK SHALL BE FABRICATED PRIOR TO RECEIPT OF APPROVED SHOP DRAWINGS.
17.	FIRE DAMPERS SHALL BE PROVIDED IN ALL RATED WALL AND CEILING PENETRATIONS PER LOCAL CODE REQUIREMENTS.
18.	GUESTROOM HVAC AND LIGHTING TO BE CONTROLLED WITH A PASSIVE INFRARED OCCUPANCY SENSOR WITH WIRELESS INTERFACE TO THE ROOM AUTOMATION SYSTEM. EXAMPLE VENDORS - INNOCOM, VERDANT & TELKONET - EACH HAVE THERMOSTATS WITH OCCUPANCY SENSORS AS WELL AS REMOTE OCCUPANCY SENSORS THAT CONTROL LIGHTING. REMOTE THERMOSTAT MUST BE HARD WIRED FOR POWER, BATTERY POWERED THERMOSTATS ARE NOT PERMITTED. WIRELESS COMMUNICATION IS ALLOWED.

MAKE-UP AIR SCHEDULE														*	*	*
TAG	TRANE BASIS OF TRANE	TOTAL CFM	MIN O.A. CFM	MAX. FAN HP	E.S.P.	MBH TOT. COOL	MBH SEN. COOL	MBH TOT. HEAT	EAT DB COOL	EAT WB COOL	TMP LEAVE COOL	WEIGHT (LB)	ELECTRICAL V/C/HZ	MCA	MOCP	NOTES
MUA-1	OADG200C1-DAB10AH00	3040	3040	2	0.5	240	97.82	300	86.2	76.8	55 DB	1870#	208/3/60	85	100	A,B,C,D
MUA-2	OADG200C1-DAB10AH00	3050	3050	2	0.5	240	97.82	300	86.2	76.8	55 DB	1870#	208/3/60	85	100	A,B,C,D
NOTES: A. SEE MAKE-UP AIR NOTES AND REQUIREMENTS - THIS SHEET B. MAKE UP AIR UNIT TO HAVE 100% FRESH AIR. C. SMOKE DETECTOR TO BE INSTALLED IN RETURN D. 12" CURB AND UNIT TO HAVE SIDE OR END DISCHARGED AND RETURN																
																* VERIFY

- Remarks:
- Factory assembled, piped, wired and tested as a single package
 - Unit shall include a 100% outside air hood with 2 position motorized damper
 - Unit shall include dual circuit cooling, 2 compressor circuits with interlocked circuit DX coil (Horizontal split not acceptable)
 - Unit shall include modulating gas heating with stainless steel heat exchanger with a 25 year warranty (parts only)
 - Unit shall include hot gas bypass on first circuit for coil frost protection at less than ambient design conditions. (Frost-stat compressor cycling not acceptable)
 - Unit shall include MODULATING hot gas reheat coil for dehumidification (2-position or dedicated heat pump circuit not acceptable)
 - Unit shall include sight glass and stainless steel drain pan
 - Unit shall include manual reset high pressure switches & auto reset low pressure switches
 - Unit shall include Condenser fan cycling control
 - Unit shall include 2 inch, 30% pleated filters
 - Control: J-Wattmaster/Orion VCM controller with ambient dewpoint sensor; electronic sequencing of compressors and heating and modulating hot gas re-heating. The intention of this unit is to provide continuous dehumidification of outside air while also providing temperature control to the space it serves If this causes overcooling in the space, the modulating hot gas reheat valve shall open to satisfy the conditioned space requirement. Field mounted controls shall include a duct mounted leaving air stat and a wall mounted stat (see plans for location). All unit mounted controls for complete operation shall be installed by the equipment manufacturer.
 - Integral non-fused disconnect switch
 - Double wall construction
 - R410a
 - Unit shall include a 30" high factory assembled and insulated roof curb/plenum. Include vibration isolation rail system model CMAB by Mason Industries
 - Compressors shall have 5 year warranty (parts only)
 - Provide MERV-8 prefilters and MERV-13 secondary filters at all air handling units exceeding 2000 cfm supply air. Provide fully synthetic media with special supporting structure to prevent moisture laden filter collapse.
 - Size ventilation air units based on ASHRAE 0.4% dehumidification data (DB/1WB 86.2° F/76.8° F to provide the required cooling coil leaving air temperature of 55°F. Heating design condition should be taken as the ASHRAE annual extreme daily mean dry bulb (1DB, -48°) Revise to meet required capacity.

- NOTES:
- ALL SUPPLY, RETURN, AND EXHAUST AIR GRILLES & DIFFUSERS TO BE PROVIDED WITH A BALANCING DAMPER.
 - ALL SUPPLY, RETURN, AND EXHAUST AIR GRILLES & DIFFUSERS TO BE PROVIDED WITH R-6 FOIL BACKED INSULATION.
 - ALL SQUARE NECKED EXHAUST AIR GRILLES TO BE PROVIDED WITH AN INSULATED ALUMINUM SQUARE TO ROUND TRANSITION.
 - LOCATIONS OF ALL SUPPLY, RETURN, AND EXHAUST GRILLES & DIFFUSERS TO BE COORDINATED WITH GENERAL, ELECTRICAL, AND FIRE PROTECTION CONTRACTORS PRIOR TO INSTALLATION.

POOL ROOM HVAC																	
TAG	DRY-O-TRON OR EQUAL DEHUMIDIFICATION	TOTAL CFM	MIN O.A. CFM	MAX FAN HP	E.S.P.	COOLING		HEATING		COMPRESSOR		WEIGHT (LB)	ELECTRICAL V/C/HZ	* MCA	* MOCP	NOTES	
						SENS.	TOTAL	INPUT	OUTPUT	QTY.	RLA	LRA					
DHL-1	DSV-030	1600	705	2	1.0	45	72	150	120	1	19	123	950	208/3/60	37	50	A,B,C
NOTES:														VERIFY			
A. EQUAL TO DRY-O-TRON MODEL DSV-030, 2000 CFM, 2HP, 208-3-60, 32 MIN. AMP, 50 AMP FUSE (VERIFY ELECTRICAL SPACE, STAT, AND HUMIDISTAT), 1 CONDENSER AND DRAIN MOUNTED ON FLOOR. INT: 950 LB WITH REHEAT AND POOL WATER HEAT OPTION, VERIFY GAS REQUIREMENTS, INSTALL PER MANUFACTURER'S SPECS.																	
B. STAINLESS STEEL HEAT EXCHANGE																	
C. PROVIDE MERV-13 FILTER																	

GRILLE / LOUVER / DIFFUSER SCHEDULES								
MARK	TYPE	MANUFACTURER	SIZE	MODEL NUMBER	MOUNTING	MATERIAL	FINISH	NOTES
GRILLE SCHEDULE								
GR-1	EXHAUST FAN	STANBRIOL	6'	-	WALL	STAINLESS STL	PAINTED	PAINT TO MATCH ADJ. SURFACE
GR-2	EXHAUST FAN	FAN TECH OR EQ	-	14x10 ALUM EXHAUST W/ LOUVERS	WALL	-	ALUM.	PAINT TO MATCH ADJ. SURFACE
OSA	FRESH AIR	GREENHECK	6' X 10'	-	WALL	-	-	-
LOUVER SCHEDULE								
LV-1	LAUNDRY	TITUS	41" x 75"	350	WALL MOUNT	ALUM	PAINT	MOUNT AT 12" A.F.F.
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-
DIFFUSER SCHEDULE								
SAG-1	S.A.G.	TITUS	24X24	301FL	CL'G.	ALUM	PAINT WHITE	-
SAG-2	S.A.G.	TITUS	6' X 60"	301	CL'G	ALUM	PAINT WHITE	-
-	-	-	-	-	-	-	-	-
RAG-1	R.A.G.	TITUS	24 X 24	350-FL	CL'G	ALUM	PAINT WHITE	-
RAG-2	R.A.G.	TITUS	6' X 60"	350	CL'G	ALUM	PAINT WHITE	-
NOTES: CL'G - CEILING MOUN S.A.G. - SUPPLY AIR GRILLE R.A.G. - RETURN AIR GRILLE								

DUCTLESS SPLIT SYSTEM SCHEDULE								
MARK	TOTAL COOLING MBH	TOTAL HEATING MBH	MINIMUM SEER	VOLTS/C/HZ	MCA	MOCP	MITSUBISHI BASIS OF DESIGN	NOTES
ACC-1	18	19	13	208/1/60	15	20	PKA-A18HA7 PLY-A18NKAY	A
ACC-2	18	19	13	208/1/60	15	20	PKA-A18HA7 PLY-A18NKAY	A
NOTES: A. 18,000 BTUH COOLING; 19,000 BTUH HEATING DUCTLESS SPLIT SYSTEM; INDOOR UNIT MAX FUSE SIZE 20 AMPS, MIN CAPACITY 15 AMP @ 208/230V/60/1C; OUTDOOR UNIT MAX FUSE SIZE 20 AMPS, MIN. CAPACITY 16 AMPS 208/230V/60/1C; PROVIDE AUTO CHANGE OVER THERMOSTAT, LOW AMBIENT OPERATION DOWN TO 0 F, AND WIND BAFFLE.								

GRB - POWER EXHAUST UNITS							
MARK	MANUFACTURER	THROAT				GREENHECK BASIS OF DESIGN	NOTES
GRB-1	GREENHECK	14 1/2"				G-095	A
NOTES: A. HOODED ALUMINUM ROOF MOUNTED EXHAUST FAN; 14 GA. MARINE ALLOY ALUMINUM WITH 1/2" MESH BIRDSCREEN; PROVIDE WITH ROOF CURB							

DUCTLED MINI-SPLIT SYSTEM SCHEDULE								
MARK	TOTAL COOLING MBH	TOTAL HEATING MBH	MINIMUM SEER	VOLTS/C/HZ	MCA	MOCP	MITSHUBISHI BASIS OF DESIGN	NOTES
DMS-1	18	18	13	208/1/60	15	20	PEAD-A18HA7 PLY-A18NKAY	A
NOTES: A. 18,000 BTUH COOLING; 18,000 BTUH HEATING DUCTLESS SPLIT SYSTEM; INDOOR UNIT MAX FUSE SIZE 20 AMPS; MIN AMPACITY 15 AMP @ 208/230V/60/1C; OUTDOOR UNIT MAX FUSE SIZE 20 AMPS; MIN. CAPACITY 16 AMPS 208/230V/60/1C; PROVIDE AUTO CHANGE OVER THERMOSTAT, LOW AMBIENT OPERATION DOWN TO 0 F, AND WIND BAFFLE.								

THRU-WALL HEAT PUMP UNITS												
MARK	SERVICE	EER/COP	COOLING CAPACITY		HEATING CAPACITY			ELECTRICAL			MODEL	NOTES
			TOTAL (BTUH)	SENSIBLE (BTUH)	AUX. HEAT KW.	TOTAL (BTUH)	EAT (°F)	VOLT/C	MCA	MOCP		
PTAC-1	GUEST ROOM'S	12/3,4	7,600	5,600	2.1	6,800	65	208/1	10.1	15	AMANA: PTH073G25AQ	①②④
PTAC-2	GUEST ROOM'S	12/3,4	8,900	6,497	3	8,100	65	208/1	14.1	20	AMANA: PTH093G35AQ	①②④

- FURNISH W/ AUXILIARY DRAIN KIT, INNOCOM E528 REMOTE THERMOSTAT & PASSIVE INFRARED OCCUPANCY SENSOR
- PROVIDE ELECTRICAL SUBBASE W/ FUSED DISCONNECT FOR PERMANENT CONNECTION.
- PROVIDE PTAC UNIT WITH INTEGRAL THERMOSTAT.
- EXTERIOR GRILLE TO BE ARCHITECTURAL GRADE, FACTORY POWDER COATED, COLOR TO MATCH BUILDING EXTERIOR.

EXHAUST FAN SCHEDULE												
MARK	SERVICE	TYPE	CFM	ESP IN. W.G.	MAX. RPM	MAX. WATTS	DRIVE	MAX. SONES	CONTROLLED BY	MODEL	NOTES	
EF-A	GUEST ROOM TOILETS	CEILING	50	0.25	875	11	DIRECT	1.0	OCCUPANCY/ MOTION SENSOR	PANASONIC: FV-0BV51	①	⑦
EF-B	WATER & ELEV EQUIP ROOMS	CEILING	228	0.25	1000	83	DIRECT	3.4	THERMOSTAT	GREENHECK: SP-A250	②	
EF-C	PUBLIC TOILETS	CEILING	80	0.25	875	11	DIRECT	1.5	SENSOR	PANASONIC: FV-0BV51	③	⑥
EF-D	FOOD PREP ICE ROOM	CEILING	105	0.25	1100	53	DIRECT	1.9	WALL SWITCH	GREENHECK: SP-A125	②	
EF-E	FITNESS AND GUEST LAUNDRY ROOMS	CEILING	105	0.25	1100	53	DIRECT	1.9	THERMOSTAT/ SENSOR	GREENHECK: SP-A125	①	②
EF-F	ELECTRIC & NETWORK ROOMS	CEILING	197	0.25	900	48	DIRECT	3.0	THERMOSTAT	GREENHECK: SP-A200	②	⑤
EF-PE	POOL EQUIPMENT ROOM	CEILING	228	0.25	1000	83	DIRECT	3.4	THERMOSTAT/ SWITCH	GREENHECK: SP-A250	②	④
EF-P	POOL ROOM	CEILING	700	0.25	1000	83	DIRECT	3.4	THERMOSTAT/ SWITCH	GREENHECK: LD/LDP-100-B	PROVIDE CURB	

- PROVIDE WITH MOUNTED ON UNIT CASING, WALL SWITCH.
- PROVIDE WITH MOUNTED ON UNIT CASING & THERMOSTAT AND WALL SWITCH.
- PROVIDE WITH MOUNTED ON UNIT CASING & INTERLOCK WITH TOILET ROOM LIGHTS.
- PROVIDE WITH MOUNTED ON UNIT CASING & THERMOSTAT AND MOTION SENSOR.
- EF-F IN PBX ROOM SET TEMPERATURE SETPOINT 2-4° ABOVE THE TEMPERATURE SETPOINT OF ACC-1 SERVING THE PBX ROOM. COORDINATE SETPOINTS WITH PBX EQUIPMENT SUPPLIERS. NOTE ACC-1 AND EXHAUST SHOULD NOT RUN AT THE SAME TIME
- AT PUBLIC RESTROOM EXHAUST FAN IS CONSTANTLY ON, AND NOT SWITCHED WITH THE LIGHTS.
- GUEST BATHROOM EXHAUST FANS MUST BE CONTROLLED BY OCCUPANCY SENSOR. GUEST BATHROOM EXHAUST FANS MUST NOT EXCEED 1.0 SONE NOISE.

FURNACE SCHEDULE											
TAG	TRANE BASIS OF DESIGN	TOTAL CFM	MAX. FAN HP	E.S.P.	MBH TOT. COOL	MBH TOT. HEAT	WEIGHT (LB)	ELECTRICAL V/C/HZ	MCA	MOCP	NOTES
FURN-1	59V2B080U5P5AA	900	1	0.5	36	80	139	115/60/1	13.9	15	A thru F
FURN-2	59V2D120U5P5AA	1760	1	0.5	60	120	156	115/60/1	13.9	15	A thru F
FURN-3	59V2D120U5P5AA	1550	1	0.5	60	120	156	115/60/1	13.9	15	A thru F
FURN-4	59V2B040U3P5AA	655	1/2	0.5	24	40	114	115/60/1	7.9	15	A thru F
FURN-5	59V2B040U3P5AA	300	1/2	0.5	24	40	114	115/60/1	7.9	15	A thru F
NOTES: A. COOLING CAPACITY BASED ON 80°F, DB/67°F, 1-WB INDOOR ENTERING AIR TEMPERATURE AND 95°F DB, ENTERING OUTDOOR TEMP. B. ESP DOES NOT INCLUDE COIL, CASING, OR FILTER LOSSES. C. PROVIDE WITH SINGLE POINT WIRING KIT D. PROVIDE EASILY ACCESSIBLE FILTER IN COMMON RETURN & OUTDOOR AIR PATH. E. THERMOSTAT - DEBONAIR COMMERCIAL PROGRAMMABLE THERMOSTAT 33C5250-F5. F. PROVIDE MERV-13 FILTER											
NOTE: ALL UNITS ARE TO BE INSTALLED WITH A DRAIN PAN WITH CONNECTION TO SANITARY. NOTE: ALL UNITS THAT ARE HUNG FROM STRUCTURE TO BE INSTALLED WITH ISOLATION DAMPERS.											

CONDENSOR SCHEDULE							
MARK	TOTAL COOLING MBH	MINIMUM SEER	VOLTS/C/HZ	MCA	MOCP	TRANE BASIS OF DESIGN	NOTES
CU-1	36.0	19	208/1/60	19	30	4TTB6036A	A,B
CU-2	60.0	19	208/1/60	34	60	4TTB6060A	A,B
CU-3	60.0	19	208/1/60	34	60	4TTB6060A	A,B
CU-4	24.0	19	208/1/60	9	15	4TTB6024A	A,B
CU-5	24.0	19	208/1/60	9	15	4TTB6024A	A,B
NOTES:							
A. ALL COMPRESSORS SHALL BE PROVIDED WITH BALL BEARING FAN MOTOR, COMPRESSOR START ASSIST, CRANKCASE HEATER, CYCLE PROTECTOR, FILTER DRYER, HIGH PRESSURE SWITCH, ISOLATION RELAY, LOW AMBIENT PRESSURE SWITCH, OUTDOOR THERMOSTAT, THERMOSTATIC EXPANSION VALVE AND TIME DELAY RELAY.							
B. REFRIGERANT PIPING SHALL NOT BE INSTALLED UNDERGROUND.							
SELECTIONS ARE BASED ON PRODUCTS BY TRANE EQUAL PRODUCTS: LENNOX, CARRIER & YORK.							

ELECTRIC WALL HEATER SCHEDULE								
MARK	SERVICE	TYPE	KW	STEPS	VOLTS/PHASE/HZ	FLA	MODEL	NOTES
EW-H-A	SEE PLANS	WALL	3	1	208/1/60	14.4	MARKEL 3450	①